FENTANYL

High-alert drug
Controlled drug

CHEMICAL NAME N-phenyl-N-[1-(2-phenylethyl)piperidin-4-yl]propanamide (as citrate)
CLASS Synthetic strong opioid agonist (phenylpiperidine); ~100× morphine potency
TRADE NAMES Sublimaze (injection) · Durogesic (patch) · Actiq · Abstral

THERAPEUTIC USES / INDICATIONS

- Analgesia during anaesthesia and in ICU (infusion).
- Severe acute pain; procedural sedation.
- Chronic stable cancer pain (transdermal patch) in opioid-tolerant patients.
- Breakthrough cancer pain (buccal/sublingual/nasal).

PHARMACODYNAMICS

- Potent **μ-opioid receptor** agonist → analgesia, sedation, respiratory depression.
- Little histamine release → more haemodynamic stability than morphine.
- High doses: muscle (chest-wall) rigidity.

ADMINISTRATION

- **IV:** bolus slowly over 1–2 min (rapid injection → chest-wall rigidity) or continuous infusion.
- **Transdermal patch:** apply to clean, dry, hairless skin; change every 72 h; rotate sites.
- Patch: avoid heat (heating pads, fever, hot baths) — increases absorption.
- Dispose of used patches by folding sticky sides together.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, constipation.
- Drowsiness, dizziness, confusion.
- Itching, sweating; patch-site reactions.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression (may be delayed with patches).
- Bradycardia, hypotension.
- Chest-wall rigidity with rapid IV doses.
- Serotonin syndrome; dependence; fatal overdose if patch misused.

DOSAGES

- **Adult IV (analgesia):** 25–100 mcg (0.5–1 mcg/kg), repeat as needed.
- **ICU infusion:** 0.5–3 mcg/kg/h, titrated.
- **Children IV:** 1–2 mcg/kg per dose.
- **Patch:** start 12–25 mcg/h only in opioid-tolerant patients; dose from oral morphine conversion.
- Doses in micrograms — double-check units.

CONTRAINDICATIONS & CAUTIONS

- Opioid-naïve patients (patch, buccal forms).
- Acute or post-operative pain (patch).
- Acute respiratory depression; severe asthma.
- **Caution:** head injury, elderly, liver disease, bradyarrhythmias.

KEY INTERACTIONS

- **CYP3A4 inhibitors** (clarithromycin, fluconazole, ritonavir): ↑ fentanyl → respiratory depression.
- **Benzodiazepines, alcohol, sedatives:** additive depression.
- **SSRIs, MAOIs:** serotonin syndrome.

PHARMACOKINETICS

- **Onset:** IV 1–2 min; patch 12–24 h. **Duration:** IV 30–60 min; patch 72 h.
- Highly lipid-soluble; metabolised in liver by CYP3A4 (inactive metabolites).
- Excreted in urine. **t½:** IV 2–4 h; after patch removal levels fall slowly (~17 h).

NURSING MANAGEMENT

- Monitor RR, SpO_2, sedation, HR and BP — continuous monitoring with infusions.
- Keep naloxone and airway equipment ready.
- Patch: record date, time and site; remove old patch before applying new one; check for fever.
- Monitor bowel function; give laxatives.
- Controlled-drug storage, documentation and disposal.

PATIENT EDUCATION

- Do not cut the patch or expose it to heat.
- Keep patches away from children; used patches still contain drug.
- Report slow breathing, extreme sleepiness or confusion.
- Avoid alcohol and sedatives.

KEY POINTS

- ~100× more potent than morphine; doses in mcg.
- Heat increases patch absorption.
- Patch for opioid-tolerant chronic pain only — change every 72 h.
- Rapid IV → chest-wall rigidity.